

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics III

Topics in Analysis (3TN)

Hilbert Spaces II: The orthogonal projection

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0.1 Orthogonal projection

In this section, we present first the notion of projection onto a non-empty subset and then the notion of **orthogonal projection onto a closed subspace**.

Definition 0.1 (Projection)

Let $(H, \cdot, \cdot)$ be a Hilbert space over the field $\mathbb{K} := \mathbb{R}$ or $\mathbb{C}$. Let F be a nonempty subset of H . Let $u \in H$. We say that a vector $u_F \in F$ is a projection of u onto F if

$$\|u - u_F\| = \inf_{v \in F} \|u - v\|. \quad (1)$$

Remark 0.2 Notice that u_F does not always exist in general. For instance, in $\mathbb{R}^2$ equipped with the Euclidean scalar product. Take F to be the open unit disc centered at the origin, that is,

$$F := \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$$

Then for every (x, y) with $x^2 + y^2 > 1$, it is clear that

$$\begin{aligned} \inf_{(x,y) \in F} [(x-a)^2 + (y-b)^2] &= \inf\{(x-a)^2 + (y-b)^2 : a^2 + b^2 < 1\} \\ &= \min\{(x-a)^2 + (y-b)^2 : a^2 + b^2 \leq 1\} \\ &= \min\{(x-a)^2 + (y-b)^2 : a^2 + b^2 = 1\} \\ &= (\sqrt{x^2 + y^2} - 1)^2. \end{aligned}$$

and is achieved through the method of the Lagrange multiplier by the vector

$$\left(\frac{x}{\sqrt{x^2 + y^2}}, \frac{y}{\sqrt{x^2 + y^2}} \right) \notin F.$$

We will consider in one the next assignments another example in an infinite dimensional spaces.

The following theorem gives a sufficient condition under which the orthogonal projection exists and is unique.

Theorem 0.3 (Orthogonal projection)

Let $(H, \cdot, \cdot)$ be a Hilbert space over the field $\mathbb{K} := \mathbb{R}$ or $\mathbb{C}$. **Let S be a closed subspace of H .** Let $u \in H$. Then there exists a unique $u_S \in S$ such that

$$\|u - u_S\| = \inf_{v \in S} \|u - v\|. \quad (2)$$

Moreover, u_S is a solution of (2) if and only if

$$\begin{cases} u_S \in S \\ u - u_S \in S^\perp \end{cases} \Leftrightarrow \begin{cases} u_S \in S \\ (u - u_S, w) = 0 \quad \forall w \in S. \end{cases} \quad (3)$$

Notation: $P_S(u) := u_S$

Proof: For the simplification, we will only prove the theorem for the case $\mathbb{K} = \mathbb{R}$. Set

$$\alpha := \inf_{v \in S} \|u - v\|.$$

So, for every $n \in \mathbb{N}^+$ there exists $v_n \in S$ such that

$$(*) \quad \alpha \leq \|u - v_n\| \leq \alpha + \frac{1}{n}.$$

Claim 1: The sequence $(v_n) \subset S$ is Cauchy in $(H, \|\cdot\|_H)$ (Banach), i.e., $\lim_{n,m \rightarrow \infty} \|v_n - v_m\| = 0$. We will make use of the parallelogram identity

$$(PI) \quad \|a + b\|^2 + \|a - b\|^2 = 2(\|a\|^2 + \|b\|^2) \quad \forall a, b \in H.$$

Take

$$a = (u - v_n)/2 \quad \text{and} \quad b = (u - v_m)/2$$

and get that

$$\|u - (v_n + v_m)/2\|^2 + \|(v_m - v_n)/2\|^2 = 2 \left[\|(u - v_n)/2\|^2 + \|(u - v_m)/2\|^2 \right].$$

Thus,

$$\begin{aligned} \|v_n - v_m\|^2 &= 2 \left[\|u - v_n\|^2 + \|u - v_m\|^2 \right] - 4 \|u - (v_n + v_m)/2\|_H^2 \\ &\leq 2 \left(\alpha + \frac{1}{n} \right)^2 + 2 \left(\alpha + \frac{1}{m} \right)^2 - 4\alpha^2 \rightarrow 0 \text{ as } n, m \rightarrow \infty. \end{aligned}$$

So, the sequence (v_n) converges to some u_S in $(H, \|\cdot\|)$. Being S closed, it follows that $v_n \in S$ and passing to the limit in (*), we get

$$\|u - u_S\| = \alpha \Leftrightarrow \|u - u_S\| = \inf_{v \in S} \|u - v\|.$$

Claim 2: The uniqueness of u_S . Assume that there is another $v_S \in S$ such $\|u - v_S\| = \alpha$. Then, applying once again the parallelogram identity (PI) this times with

$$a = (u - u_S)/2 \quad \text{and} \quad b = (u - v_S)/2$$

and get that

$$\|u - (u_S + v_S)/2\|^2 + \|(u_S - v_S)/2\|^2 = 2 \left[\|(u - u_S)/2\|^2 + \|(u - v_S)/2\|_H^2 \right].$$

Thus,

$$\begin{aligned}\|u_S - v_S\|^2 &= 2\left[\|u - u_S\|^2 + \|u - v_S\|^2\right] - 4\|u - (u_S + v_S)/2\|^2 \\ &\leq 2\alpha^2 + 2\alpha^2 - 4\alpha^2 = 0 \quad (\text{because } (u_S + v_S)/2 \in S \text{ subspace})\end{aligned}$$

Therefore, $u_S = v_S$.

Claim 3: Let us now prove (2) $\Leftrightarrow$ (3)

$(\Rightarrow)$: Assume that (2) holds and let us establish (3). Let $w \in S$. Let $\lambda \in \mathbb{R}$ and take $v = u_S \pm \lambda w \in S$ (subspace) in (2) and get

$$\begin{aligned}\|u - u_S\|^2 &\leq \|u - u_S \pm \lambda w\|^2 \\ &= \|u - u_S\|^2 \pm 2\lambda(u - u_S, w) + \lambda^2\|w\|^2\end{aligned}$$

from which we get that

$$\pm(u - u_S, w) \leq \frac{\lambda}{2}\|w\|^2 \quad \forall \lambda \in \mathbb{R} \setminus \{0\}$$

Letting $\lambda \rightarrow 0$ we get $\pm(u - u_S, w) \leq 0 \Leftrightarrow (u - u_S, w) = 0$.

$(\Leftarrow)$: Let $v \in S$ and take $w = v - u_S \in S$ in (3) and get

$$\begin{aligned}(u - u_S, v - u_S) &= 0 \Leftrightarrow (u - u_S, (u - u_S) + (v - u)) = 0 \\ \Leftrightarrow \|u - u_S\|^2 + (u - u_S, v - u) &= 0 \Leftrightarrow \|u - u_S\|^2 \leq \|u - u_S\| \|u - v\| \quad (\text{CS}).\end{aligned}$$

Therefore, we get that $\|u - u_S\| \leq \|u - v\| \quad \forall v \in S$. So, u_S satisfies (2). ■

Example 0.4 Let $\mathbb{R}^4$ be equipped with the usual Euclidean scalar product

$$(u, v) = \sum_{i=1}^4 u_i v_i \quad \forall u = (u_1, \dots, u_4), v = (v_1, \dots, v_4) \in \mathbb{R}^4.$$

Let $v_1 = (1, 0, 1, 1)$ and $v_2 = (0, -1, 0, 1)$. Set $S := \text{Span}(v_1, v_2)$.

(a) Find a basis of $S^\perp$.

(b) Let $u = (-2, 1, 1, 0)$. Show that $u \notin S$ and find $P_S(u)$ the orthogonal projection of u onto S .

Solution:

(a) By definition $w = (a, b, c, d) \in S^\perp$ if and only if

$$(w, v_1) = (w, v_2) = 0 \Leftrightarrow \begin{cases} a + c + d = 0, \\ -b + d = 0 \end{cases} \Leftrightarrow \begin{cases} a = -c - d, \\ b = d \end{cases}$$

Therefore, we write

$$(a, b, c, d) = (-c - d, d, c, d) = c(-1, 0, 1, 0) + d(-1, 1, 0, 1).$$

Hence

$$S^\perp \subset \text{Span}((-1, 0, 1, 0), (-1, 1, 0, 1)).$$

Now, notice that each of the vector $(-1, 0, 1, 0)$ and $(-1, 1, 0, 1)$ is orthogonal to both v_1 and v_2 . So, we also have

$$S^\perp \supset \text{Span}((-1, 0, 1, 0), (-1, 1, 0, 1)).$$

Therefore,

$$S^\perp = \text{Span}((-1, 0, 1, 0), (-1, 1, 0, 1)).$$

Since the two vectors $(-1, 0, 1, 0)$ and $(-1, 1, 0, 1)$ are linearly independent, we find they form a basis of $S^\perp$.

(Have seen you how we got the solution by just applying the definitions?)

- (b) Let us show that $u \notin S$. This just means that u cannot be written as a linear combination of v_1 and v_2 .

Now in order to find $P_S(u)$ we just need to find $\alpha, \beta \in \mathbb{R}$ such that

$$P_S(u) = \alpha v_1 + \beta v_2 \quad \text{and} \quad \begin{cases} (u - P_S(u), v_1) = 0, \\ (u - P_S(u), v_2) = 0 \end{cases}$$

You will obtain a system of two linear equations with two unknowns that you should be able to solve it.

Once you find α and β then you can get $P_S(u)$.

What we learnt in this example is that the formulation of $P_S(u)$ as

$$\begin{cases} P_S(u) \in S, \\ (u - P_S(u), v) = 0 \quad \forall v \in S \end{cases}$$

is very practical for finding $P_S(u)$.

We recall a well-known procedure of constructing an orthonormal system from any linearly independent system.

0.2 Gram-Schmidt orthonormalization process

Theorem 0.5 (Gram-Schmidt orthonormalization process)

Let $\{v_n\}_{n \geq 1}$ be a linearly independent system in the inner product space $(H, (\cdot, \cdot))$. Then there exists an orthonormal system $\{e_n\}_{n \geq 1}$ with the property that

$$\text{Span}\langle (v_k)_{k=1}^n \rangle = \text{Span}\langle (e_k)_{k=1}^n \rangle \quad \forall n \geq 1,$$

where given subset S of H , $\text{Span}(S)$ denotes the set of all linear combinations of vectors in S .

Proof: By induction on n . For $n = 1$ we consider v_1 and we define

$$e_1 := \frac{v_1}{\|v_1\|}.$$

Assume that for $n \geq 2$ we have constructed $\{e_i\}_{i=1}^{n-1}$, orthonormal system with the property

$$\text{Span}\langle \{v_i\}_{i=1}^{n-1} \rangle = \text{Span}\langle \{e_i\}_{i=1}^{n-1} \rangle.$$

To find the vector e_n , we first result the vector v_n as

$$v_n = u_n + w_n \quad \text{with } u_n \in \text{Span}\langle (e_i)_{i=1}^{n-1} \rangle \text{ and } w_n \in (\text{Span}\langle (e_i)_{i=1}^{n-1} \rangle)^\perp.$$

Therefore $u_n = \sum_{i=1}^{n-1} \lambda_{n,i} e_i$ and $(w_n, e_i) = 0$ for every $i = 1, 2, \dots, n-1$, and hence from the latter it follows that $\lambda_{n,i} = (v_n, e_i)$. Finally setting

$$e_n := \frac{w_n}{\|w_n\|} = \frac{v_n - \sum_{i=1}^{n-1} (v_n, e_i) e_i}{\left\| v_n - \sum_{i=1}^{n-1} (v_n, e_i) e_i \right\|}$$

we obtain the desired properties. ■

Exercise 0.6 Let $\mathbb{R}_2[X]$ be the vector space of polynomials with real coefficients and degree less than or equal 2. Consider the mapping defined by

$$(p, q) := \sum_{i=0}^3 p(i)q(i) \quad \forall p, q \in \mathbb{R}_2[X].$$

- (a) Show that $(\cdot, \cdot)$ defines a scalar product on $\mathbb{R}_2[X]$.
- (b) Find an orthonormal basis of $\mathbb{R}_2[X]$ for that scalar product.

Hint: you can start from the system $\{1, x, x^2\}$ which is a basis on $\mathbb{R}_2[X]$.

Our next lecture will be on the differentiability of functions between normed spaces!